SHORTS

WHY MARS IS THE BEST PLANET

About four billion years ago, the two planets, Earth and Mars had many things in common. Craters pockmarked their faces, and volcanoes rose from their plains. However, unlike Earth, Mars wasn't so lucky, and over the course of time, Mars dried up.

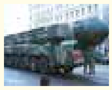
http://dgit.in/MarStudy

OLD-FASHIONED TECH IN ROGUE ONE

From switches, cables, heavy doors, grenades, a staff, and sniper rifles, to stormtroopers driving trucks rather than a zero-gravity trawler, critics explain why the technology in the new Star Wars movie looks like something from WWII.

http://dgit.in/SWtechRO

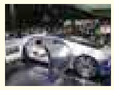
STOPPING A NUCLEAR MISSILE



North Korea is researching a variety of missiles that, with research and development, could be converted to ICBMs. Can America defend itself against an incoming ballistic missile? Are current age Lasers-armed drones and interceptor missiles good enough?

http://dgit.in/StopNuc

THE FUTURE OF CARS



Electrification, autonomous driving and new ownership models are the three new sifts that almost all the major carmakers are embracing. Here are some reasons why we should be optimistic about the future of car technology.

http://dgit.in/CarsF

What's the big deal about Jarvis, Facebook CEO's very own AI assistant?

Interactive page

Ever since the Facebook CEO published his note on Jarvis, his personal AI programmed for home, we've all been excited with the prospect of an AI interface controlling our homes. What does it mean, though, for the progress of connected AI assistants?

– By Souvik Das

Facebook's CEO Mark Zuckerberg loves coding. That's very clear from his own post, and the recent reports surrounding his latest personal project - Jarvis. At the beginning of 2016, he had written a post on how his 2016 challenge was to build a simple, artificial intelligence algorithm to run basic operations at home. On December 19, he published a note on his 100 hours of coding through the year, which he spent building Jarvis. The naming is not a mere coincidence, and Zuckerberg's intention is somewhat similar to that of Tony Stark - build an assistant that, in the truest sense, will be an intelligent, highly personal virtual helper.

Naturally, most of us have been quite excited about this recent development. But, what is all this excitement about, and what does this signify for the advancement and maturity of Artificial Intelligence and the Internet of Things? In a demonstration of IBM's Watson AI in India, Sriram Raghavan of IBM India's Research, stated, "The Internet of Things is an equally pivotal factor, and the rise in awareness and enthusiasm surrounding IoT will be crucial in taking cognitive platforms to everyday households." So how does Jarvis contribute to this?

Who's Jarvis?

Jarvis is reportedly an artificial intelligence algorithm that Zuckerberg himself has coded. He has used the vast expanse of Facebook's programming and language tools at his disposal, and has reportedly spent a cumulative duration of 100 hours in making it. He set about with the idea of building a structured algorithm that can read and receive simple instructions in voice or text, to control simple operations.

Jarvis: Not quite this, actually

For instance, "switch on bedroom lights" or "pop up the toaster".

The AI assistant can also learn personal music tastes to deliver customised playlists, recognise voices and even enable household appliances to execute certain chores. Pivotal to Jarvis' "mind" is its server - the centrepiece of the information flow. On one hand were the interfaces from which tasks were fed to Jarvis. These included a text interface integrated into Facebook's Messenger app with a custom bot, built by Zuckerberg himself. The second was voice input, via a custom iOS application coded for Jarvis. The third was direct input of photographs and video feeds connected to Jarvis, for facial recognition.

To execute tasks provided by the input interfaces, Jarvis' intelligence depended on three AI systems - a natural language processing engine to understand context and improvisation instead of preset commands, a speech recognition engine to recognise and respond to voices contextually, and a face recognition engine that will read information off a storage vat. All